

Lab 01

TITLE

**“GitHub Identity, Student Portal Registration, and
Ubuntu Server Installation”**

Name:

Areeba Hameed

Registration number:

(2023-BSCS-25)

Section:

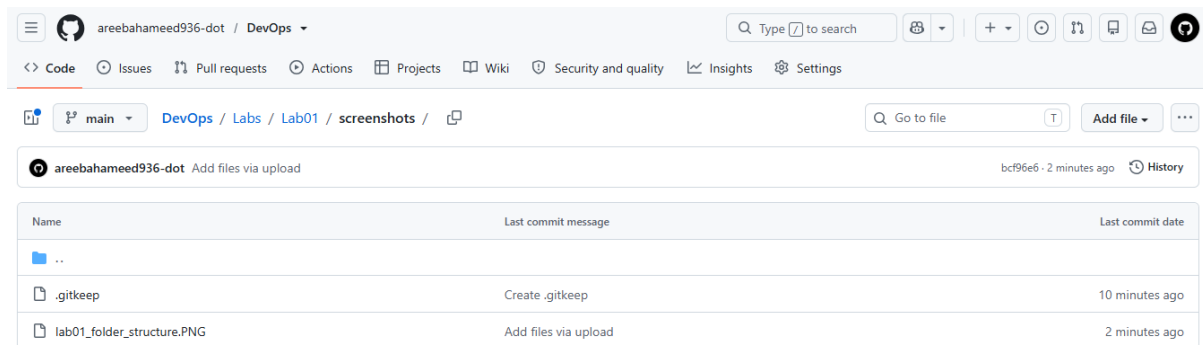
A

GitHub username:

areebahameed936-dot

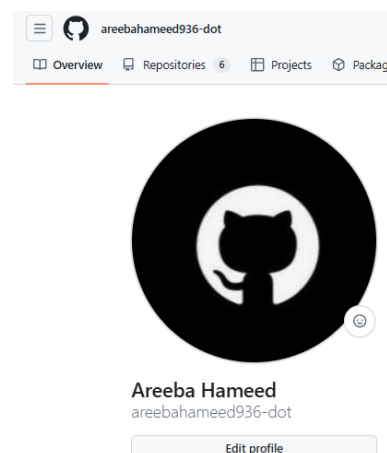
Getting Started

1. Create the following directory inside your DevOps repository:
2. Labs/Lab01/screenshots/
 - a. Save a screenshot showing the created directory structure as lab01_folder_structure.png.
3. Read the complete task list before beginning the installation.



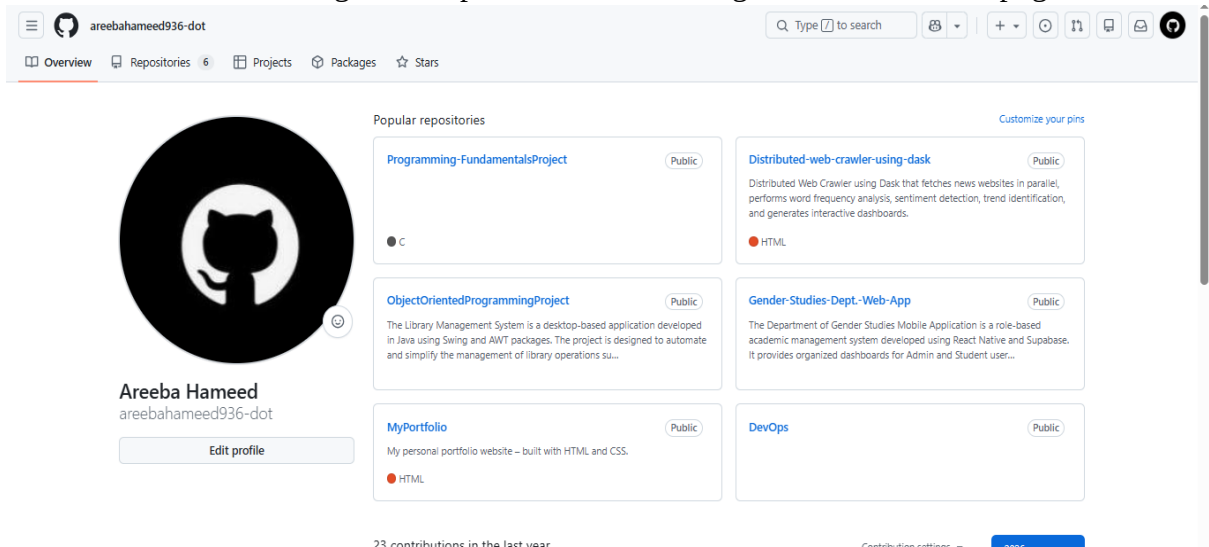
Task 1: Create and Configure Your GitHub Account

1. Open GitHub Sign Up and create your personal GitHub account.
 - Use your own email address and actual identity.
 - Do not create an account using another student's information.
2. Choose a professional GitHub username related to your actual name.
 - Do not use spaces in the GitHub username.
 - Prefer lowercase letters and numbers because the same username will be used in Ubuntu.
 - Save a screenshot clearly showing your profile name and GitHub username as github_profile.png.



3. Open GitHub > Settings > Public profile.

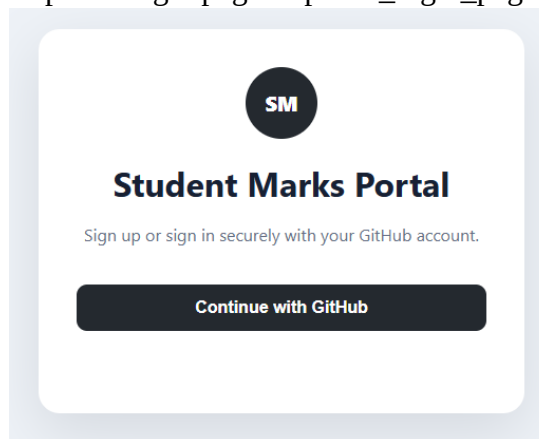
- Enter your complete actual name in the Name field.
- Save a screenshot showing the completed Name field as `github_actual_name.png`.



- Do not include your password, access token, recovery codes, or other private credentials in any screenshot.

Task 2: Register in the Student Marks Portal

1. Open the [Student Marks Portal](#).
- Save a screenshot of the portal login page as `portal_login_page.png`.



2. Select Continue with GitHub and sign in using the GitHub account created in Task
3. Enter your actual registration number using the format provided by your instructor, for example:

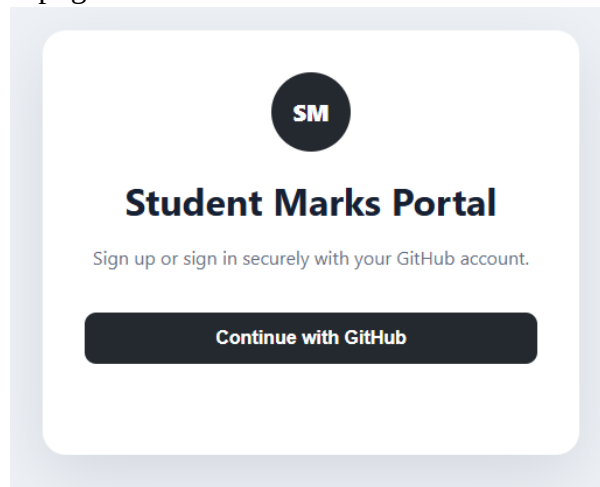
2024-BSE-00

Select the correct course code.

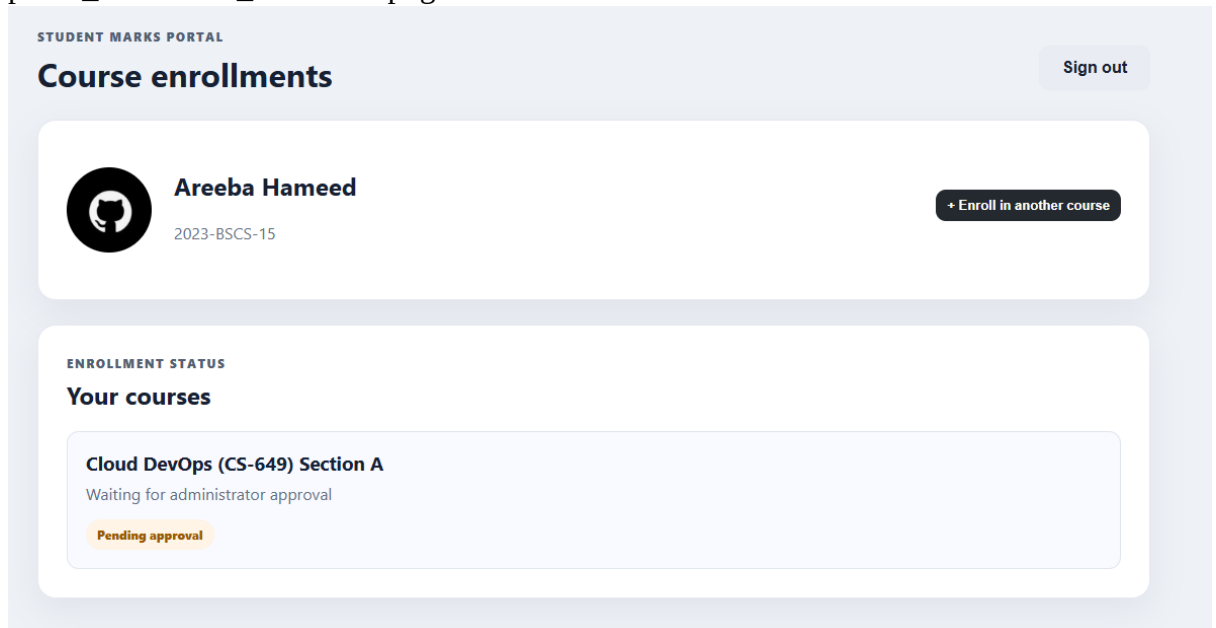
Select your actual section, either A or B.

Review your information and submit the enrollment for approval.

Save a screenshot showing your GitHub profile information inside the portal as portal_github_profile.png.



Save a screenshot of the submitted or pending enrollment as portal_enrollment_submitted.png.





Areeba Hameed

GitHub authenticated

Registration
2023-BSCS-15

GitHub username
[areebahameed936-dot](#)

Email
[areebahameed936@gmail.com](#)

Approved courses
1

[+ Enroll in another course](#)

APPROVED COURSES

Course Marks

Not published

Cloud DevOps (CS-649) Section A

ASSESSMENT	OBTAINED	TOTAL	PERCENTAGE
------------	----------	-------	------------

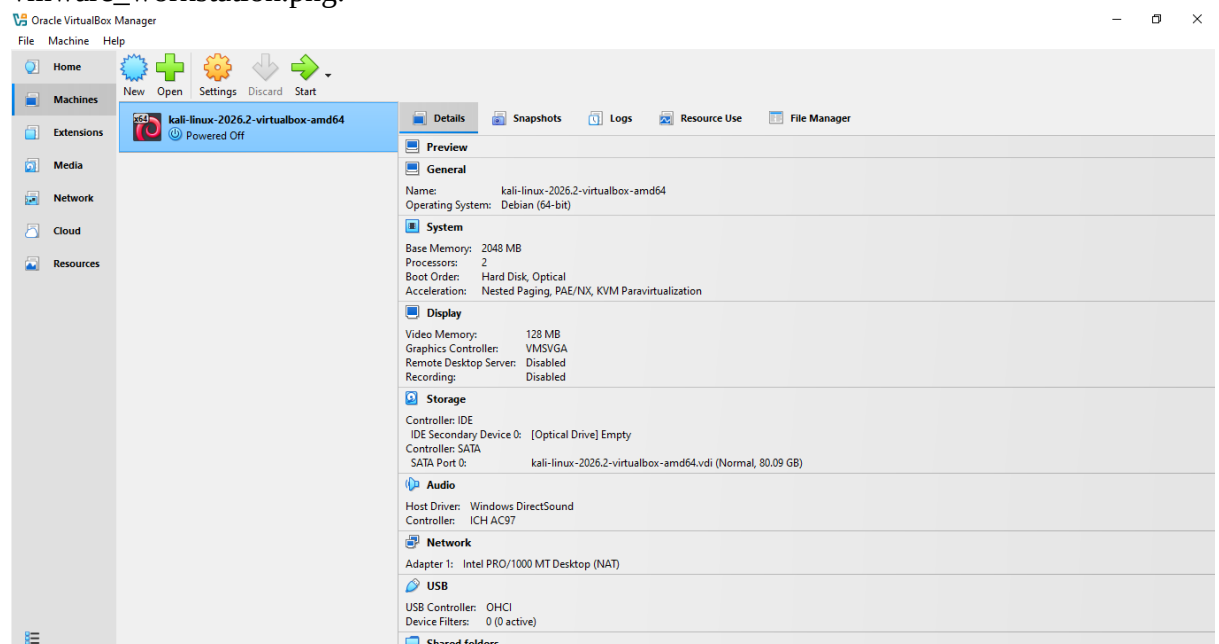
Marks have not been published.

A registration number belongs to only one student. Never enter another student's registration number.

Task 3: Download VMware Workstation Pro and Ubuntu Server

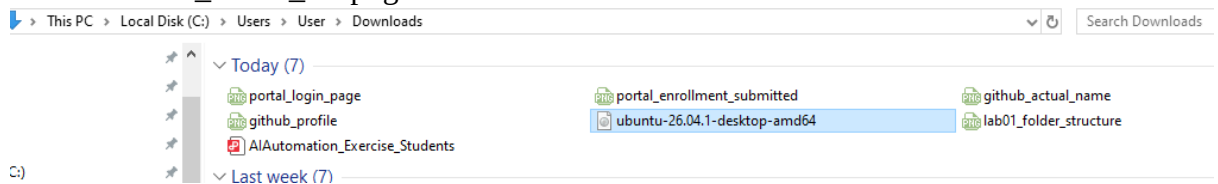
1. Download and install VMware Workstation Pro on your Windows computer.

- If VMware Workstation Pro is already installed, open it and verify that it works.
- Save a screenshot showing VMware Workstation Pro running as `vmware_workstation.png`.



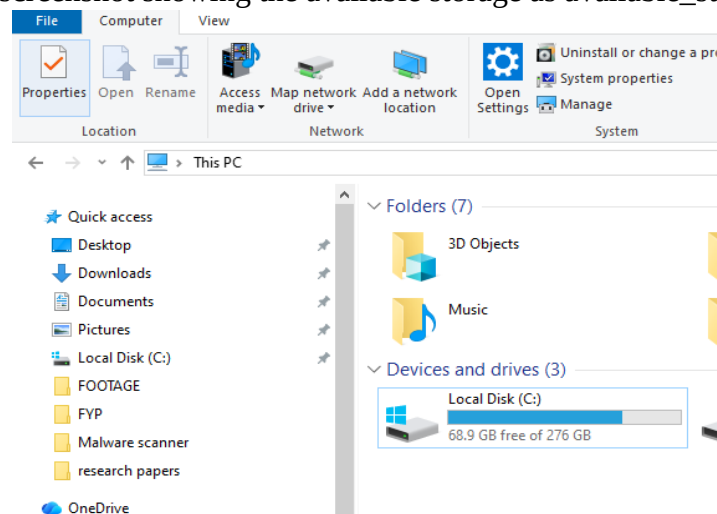
2. Open the Ubuntu Server download page and download the latest Ubuntu Server LTS ISO.

- Save a screenshot showing the downloaded Ubuntu Server ISO filename as `ubuntu_server_iso.png`.



3. Confirm that at least 20 GB of free storage is available for the Ubuntu Server virtual machine.

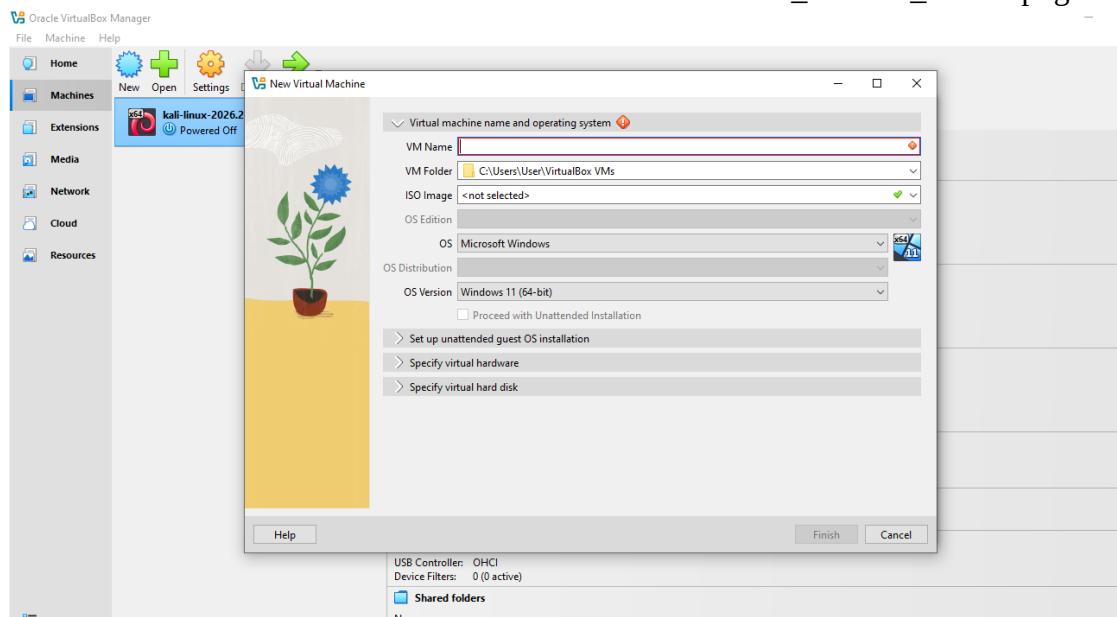
- Save a screenshot showing the available storage as `available_storage.png`.



Task 4: Create the Ubuntu Server Virtual Machine

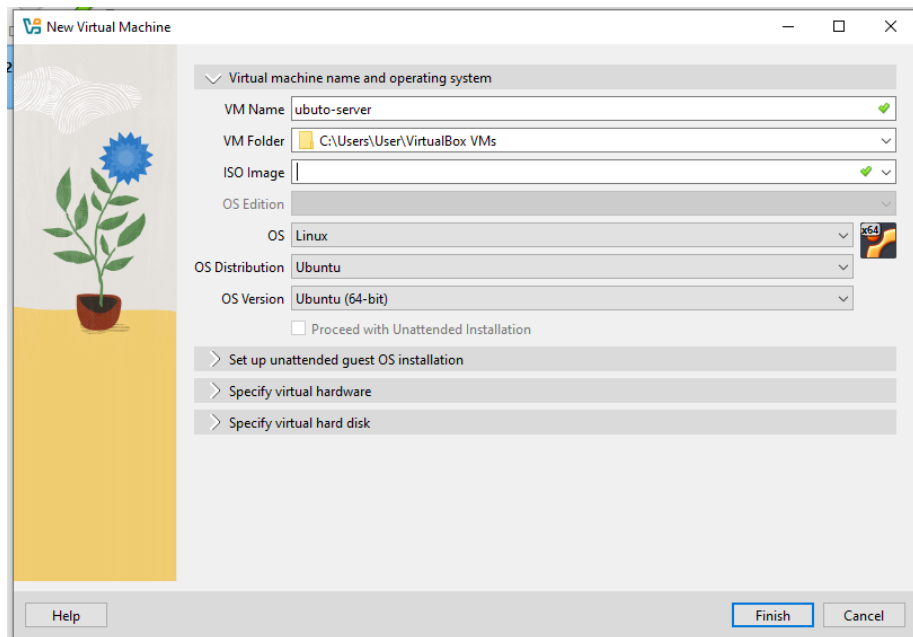
1. Open VMware Workstation Pro and select Create a New Virtual Machine.

- Save a screenshot of the New Virtual Machine Wizard as `vm_creation_wizard.png`.

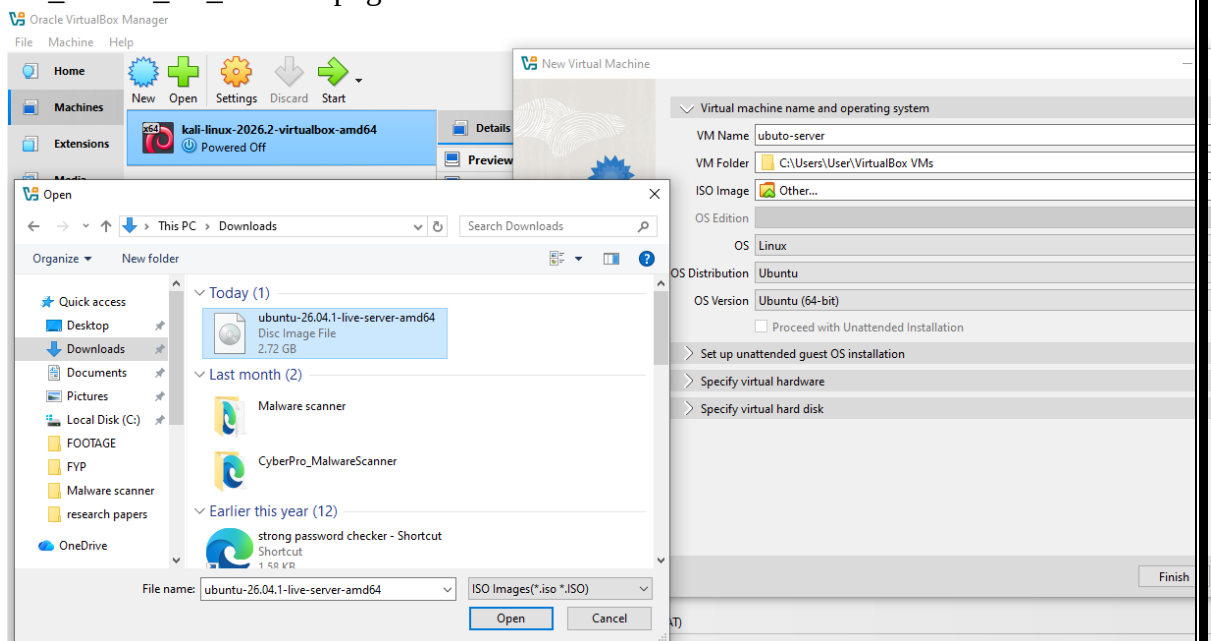


2. Select Typical (recommended) as the virtual machine configuration.

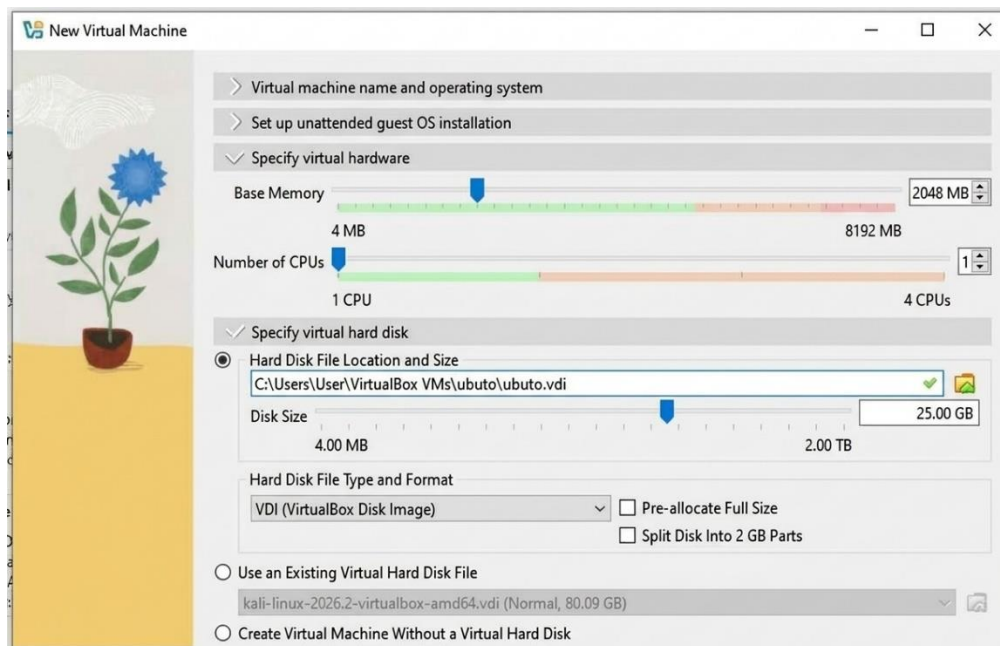
- Save a screenshot showing this selection as `vm_typical_configuration.png`.



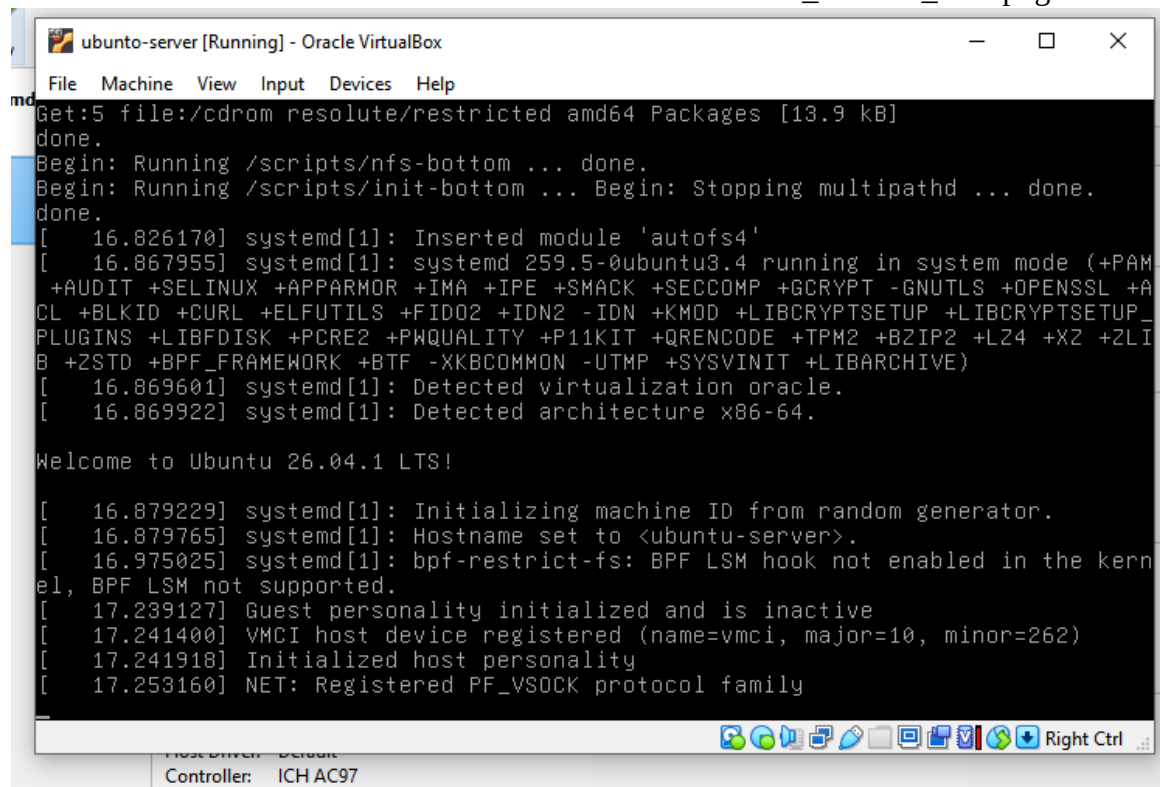
3. Select Installer disc image file (ISO) and browse to the Ubuntu Server ISO downloaded in Task 3.
 - Save a screenshot showing the selected Ubuntu ISO path as `vm_ubuntu_iso_selected.png`.



4. Allocate at least 20 GB of virtual disk space and complete the virtual machine wizard.
 - Save a screenshot showing the final virtual machine configuration as `vm_configuration_summary.png`.



5. Power on the virtual machine and wait for the Ubuntu Server installer to start.
 - Save a screenshot of the Ubuntu Server boot screen as ubuntu_installer_boot.png.

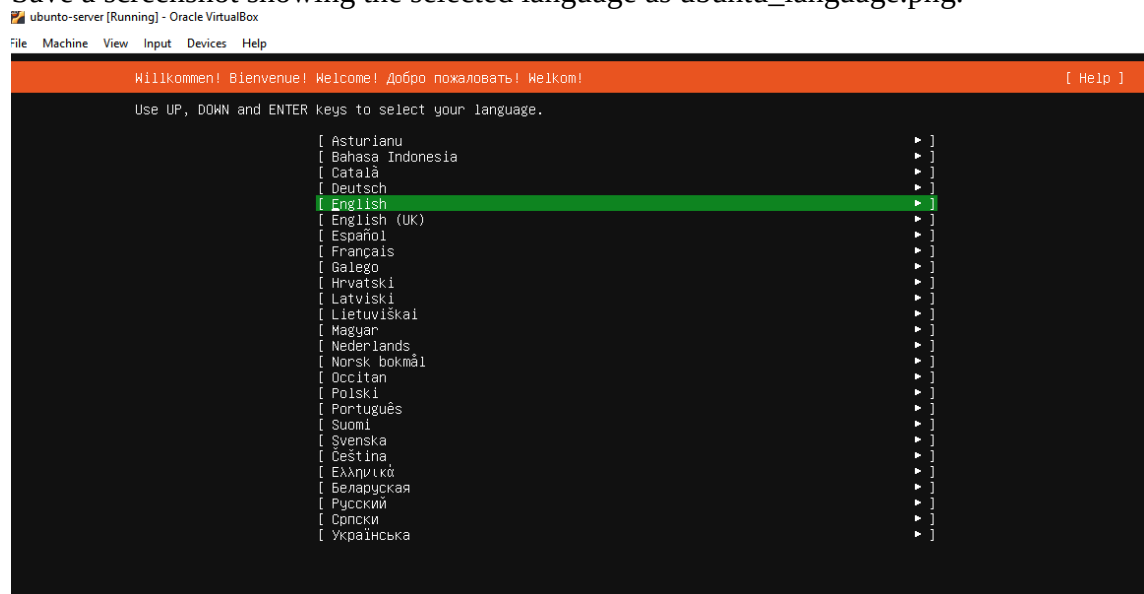


Instructional reference

The following image is provided to help you recognize the Ubuntu Server boot screen:

Task 5: Choose the Installation Language

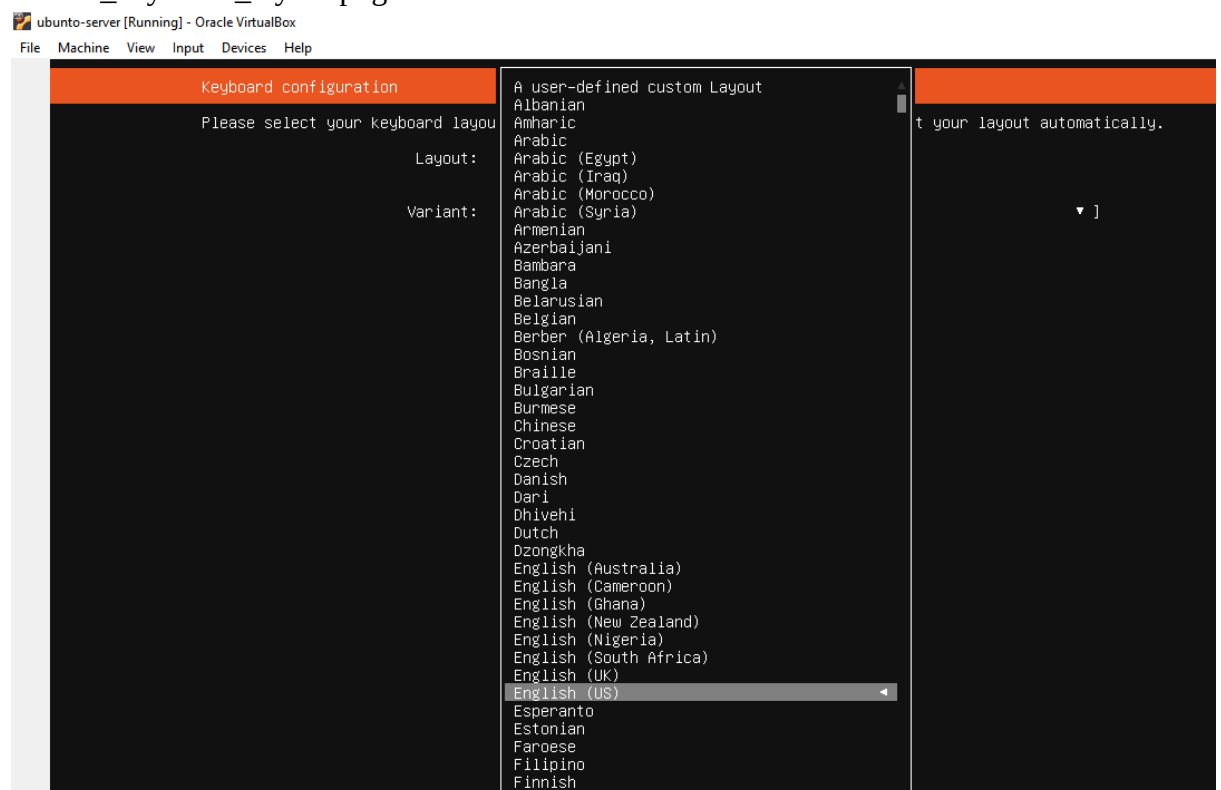
1. Use the arrow keys to select the required installation language.
- Save a screenshot showing the selected language as ubuntu_language.png.



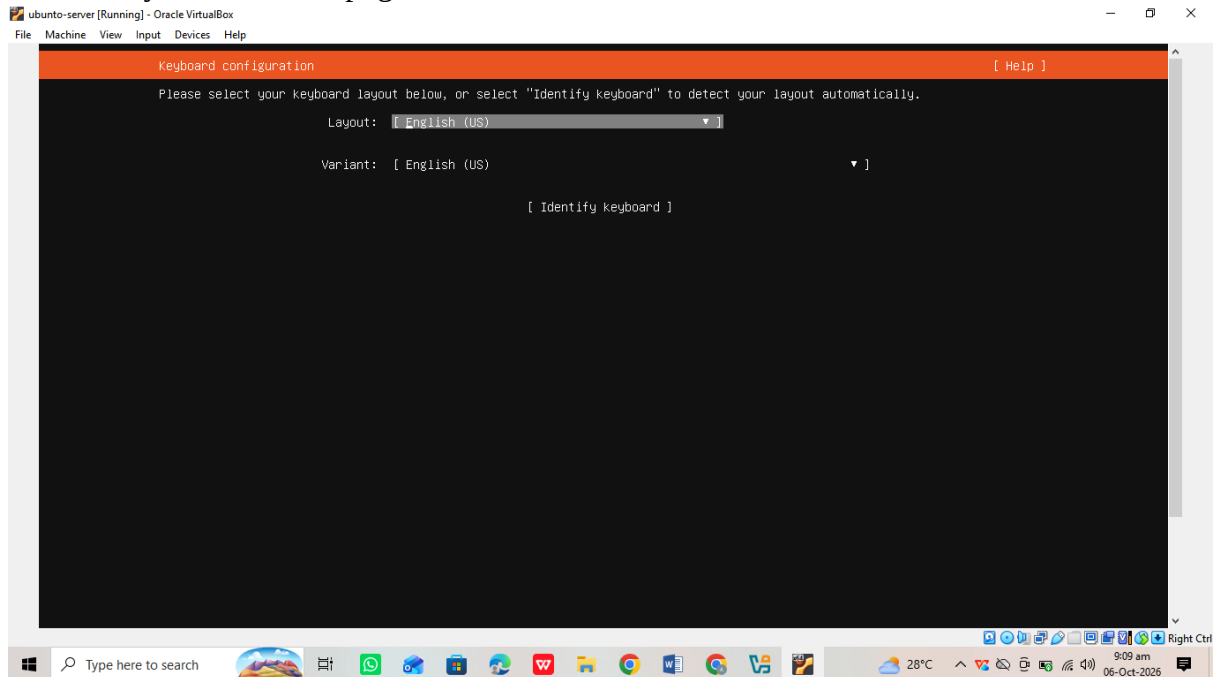
2. Press Enter to continue.

Task 6: Configure the Keyboard Layout

1. Select the keyboard layout that matches your keyboard.
- Save a screenshot showing the selected keyboard layout as ubuntu_keyboard_layout.png.



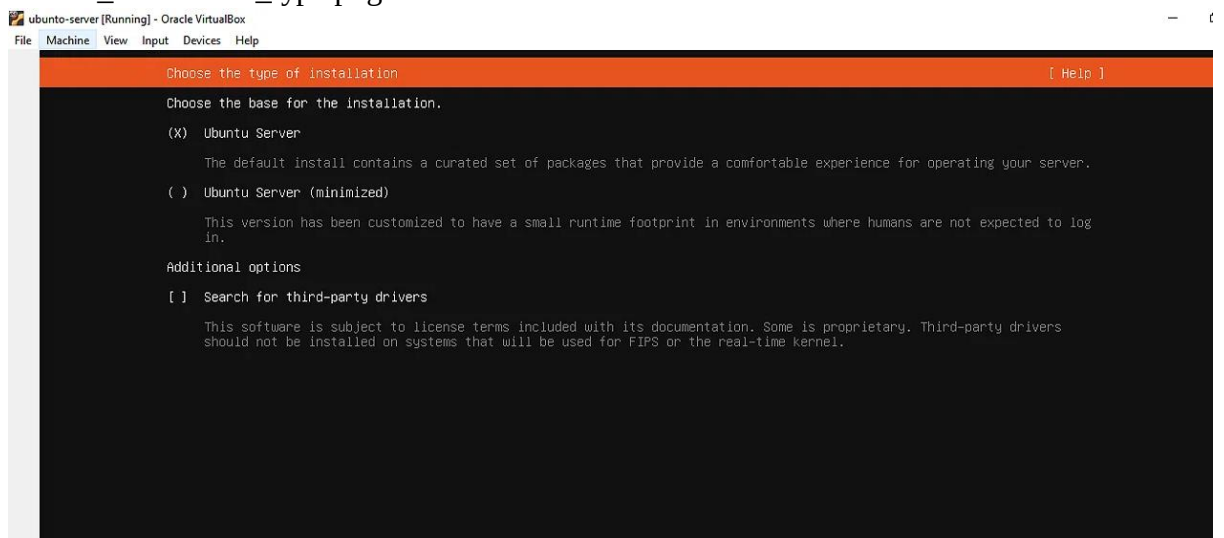
2. Select the appropriate keyboard variant.
- Save a screenshot showing the selected keyboard variant as `ubuntu_keyboard_variant.png`.



3. Select Done to continue.

Task 7: Choose the Ubuntu Server Installation Type

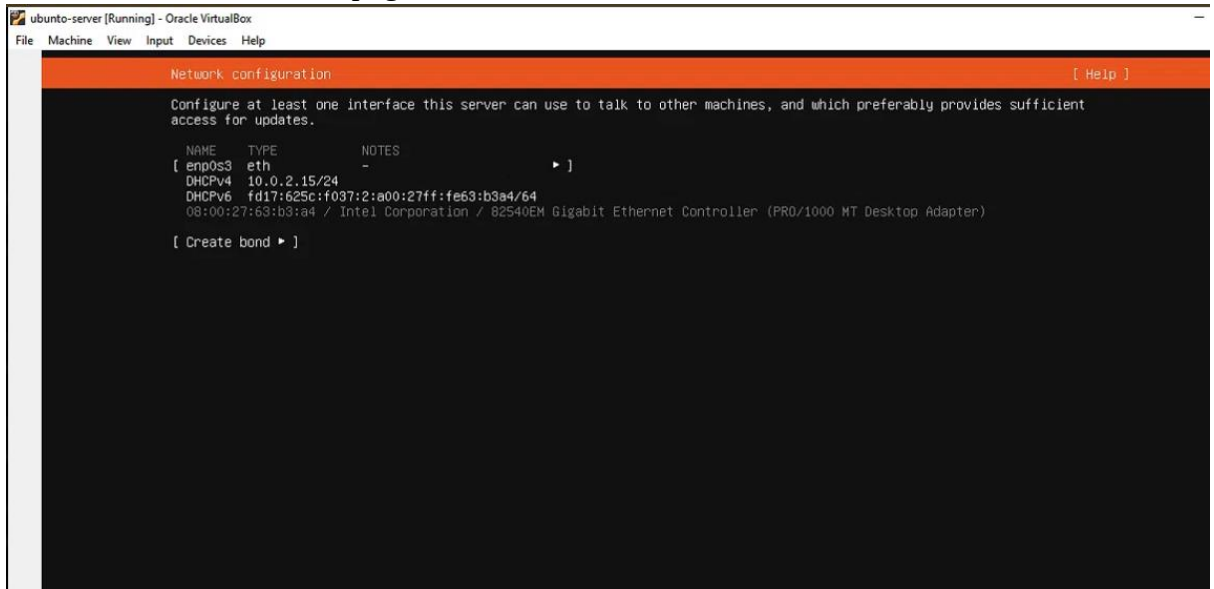
1. Select the normal Ubuntu Server installation option.
- Do not select a MAAS installation unless your instructor specifically asks for it.
- Save a screenshot showing the selected installation type as `ubuntu_installation_type.png`.



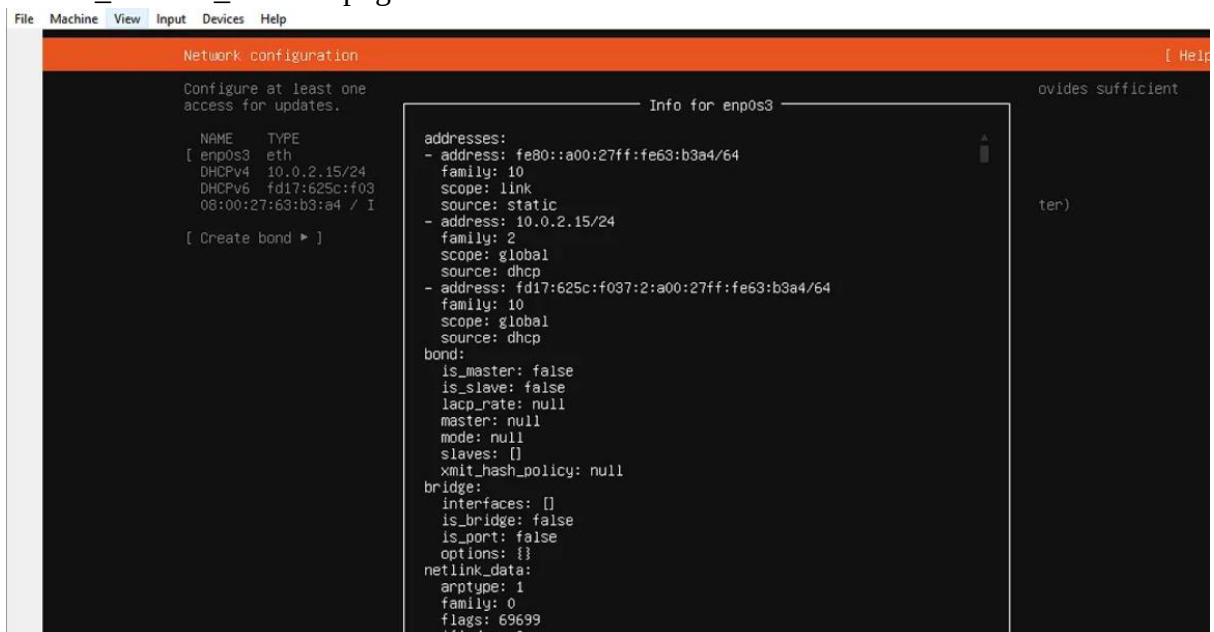
2. Continue to the next installer screen.

Task 8: Configure the Network

1. Review the network interface detected by the Ubuntu Server installer.
 - Save a screenshot showing the detected network interface as `ubuntu_network_interface.png`.



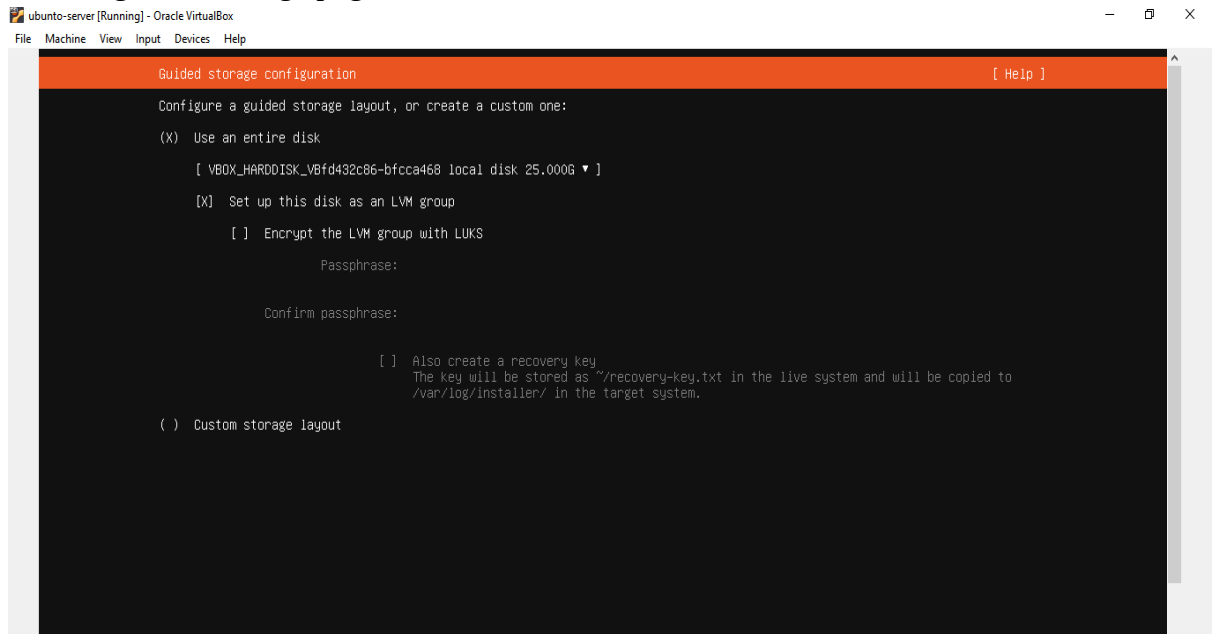
2. Confirm that the interface has received an IP address automatically through DHCP.
 - Save a screenshot showing the assigned network address as `ubuntu_installer_network.png`.



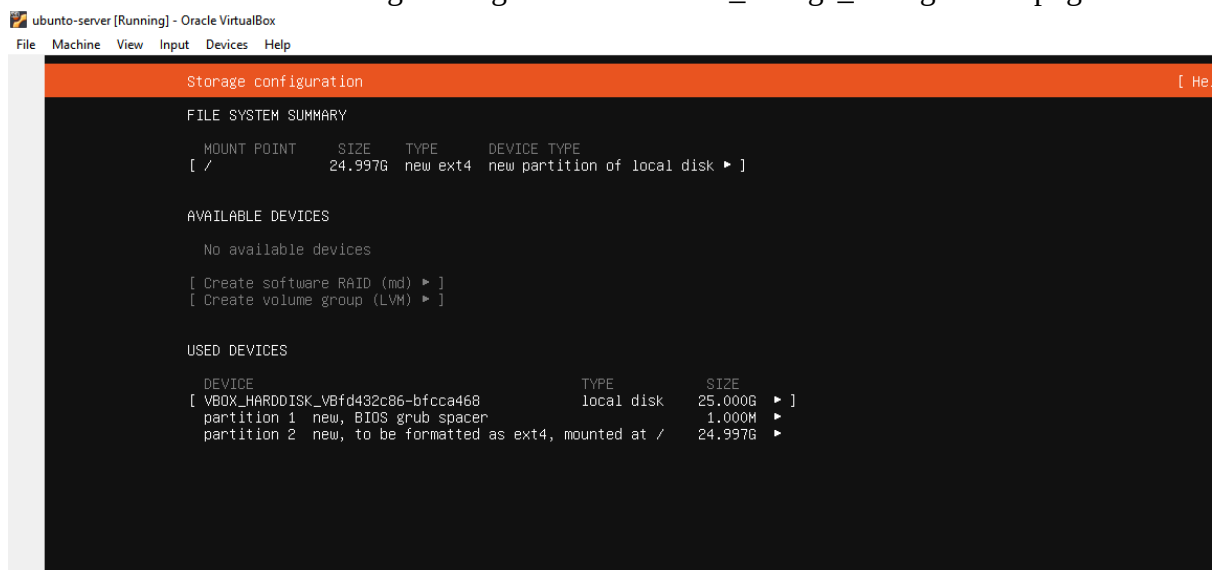
3. Continue when the network configuration is correct.

Task 9: Configure Storage

1. Select the guided storage option to use the VMware virtual disk.
- Save a screenshot showing the selected guided storage option as `ubuntu_guided_storage.png`.



2. Review the storage configuration before continuing.
- Save a screenshot of the storage configuration as `ubuntu_storage_configuration.png`.



Use only the VMware virtual disk created for this lab. Selecting or formatting the wrong physical disk can destroy existing data.

Task 10: Select the Installation Disk

1. Select the virtual disk created for the Ubuntu Server virtual machine.
- Save a screenshot showing the selected disk as `ubuntu_selected_disk.png`.

USED DEVICES

DEVICE	TYPE	SIZE	
[VBOX_HARDDISK_VBfd432c86-bfcca468	local disk	25.000G	▶]
partition 1 new, BIOS grub spacer		1.000M	▶
partition 2 new, to be formatted as ext4, mounted at /		24.997G	▶

2. Check the displayed disk size and make sure it is the correct VMware virtual disk.
- Save a screenshot showing the disk name and size as ubuntu_disk_details.png.

AVAILABLE DEVICES

No available devices

[Create software RAID (md) ▶]
[Create volume group (LVM) ▶]

USED DEVICES

DEVICE
[VBOX_HARDDISK_VBfd432c86-bfcca468
partition 1 new, BIOS
partition 2 new, to

Info for VBOX_HARDDISK_VBfd432c86-bfcca468

Path:	/dev/sda
Multipath:	unknown
Vendor:	ATA
Model:	VBOX_HARDDISK
SerialNo:	VBOX_HARDDISK_VBfd432c86-bfcca468
WMN:	unknown
Size:	26843545600
Bus:	ata
Rotational:	true
Path:	/devices/pci0000:00/0000:00:0d.0/ata3/host2/target2
	:0:0/2:0:0:0/block/sda
NVMe controller:	None

[Close]

Task 11: Review the Partitions

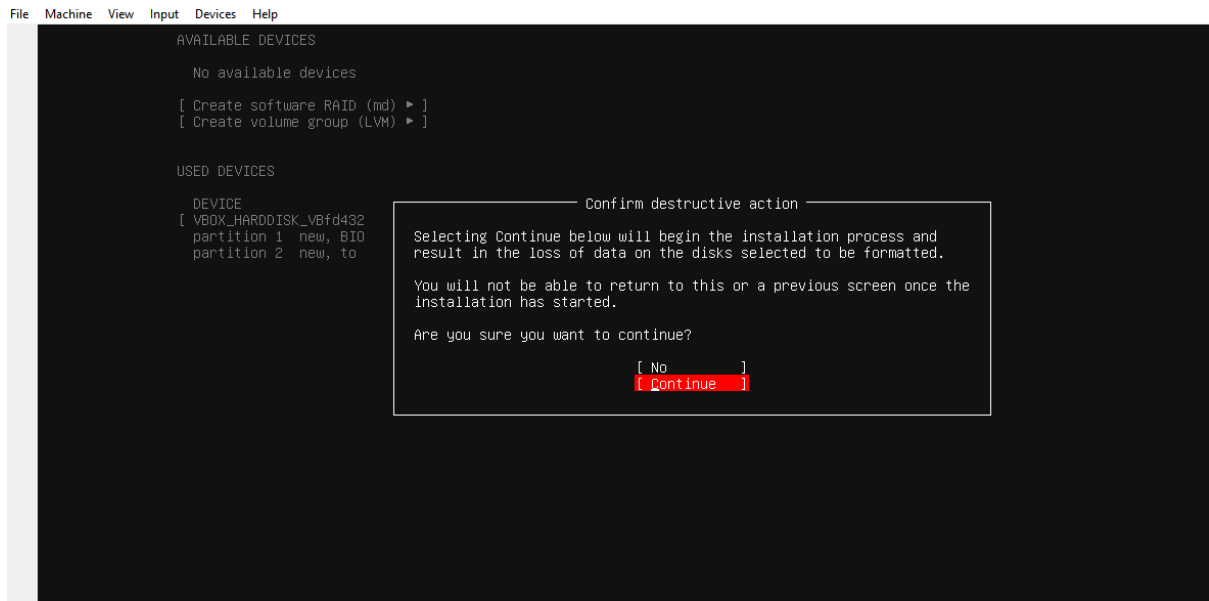
1. Review the partition layout proposed by the Ubuntu Server installer.
- Save a screenshot showing the proposed partitions as ubuntu_partition_layout.png.

[VBOX_HARDDISK_VBfd432c86-bfcca468	local disk	25.000G	▶]
partition 1 new, BIOS grub spacer		1.000M	▶
partition 2 new, to be formatted as ext4, mounted at /		24.997G	▶

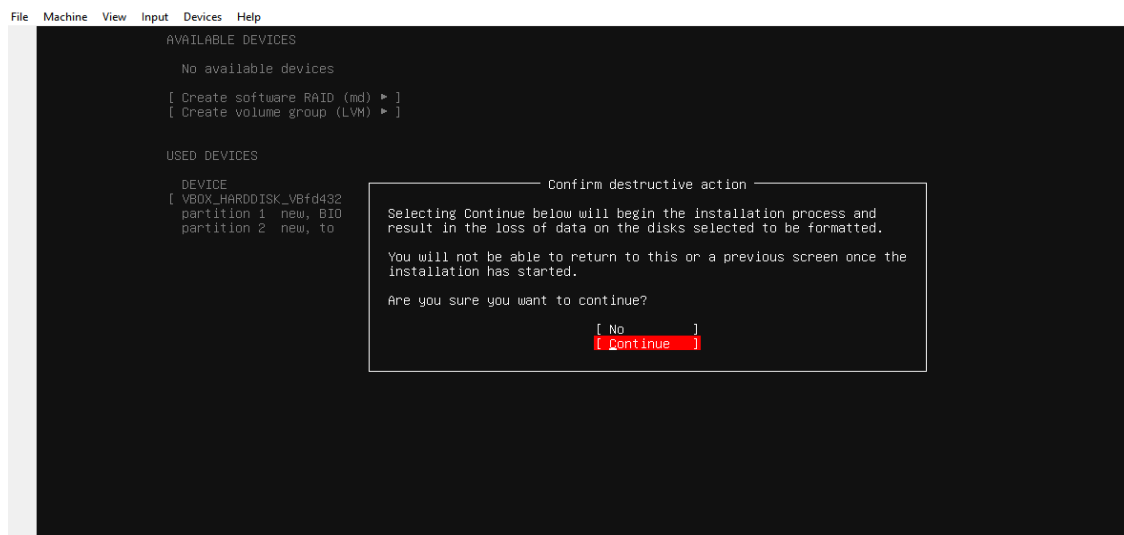
2. Confirm that the partitions belong to the VMware virtual disk.
3. Select Done only after reviewing the complete layout.

Task 12: Confirm the Storage Changes

1. Read the destructive-action warning carefully.
 - Save a screenshot showing the final warning as ubuntu_storage_warning.png.



2. Confirm the changes only when you are certain that the VMware virtual disk is selected.
 - Save a screenshot showing the confirmation selection as ubuntu_storage_confirmed.png.

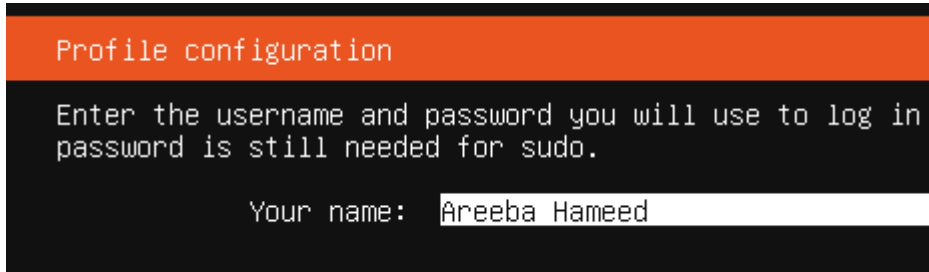


Task 13: Configure the Ubuntu Profile

Complete the Ubuntu profile using the following required values:

Installer field	Required value
Your name	Your complete actual name
Your server's name	ubuntu
Pick a username	Your exact GitHub username
Choose a password	A secure password you can remember

1. Enter your complete actual name in the Your name field.
- Save a screenshot showing the entered name as ubuntu_actual_name.png.



Profile configuration

Enter the username and password you will use to log in password is still needed for sudo.

Your name: Areeba Hameed

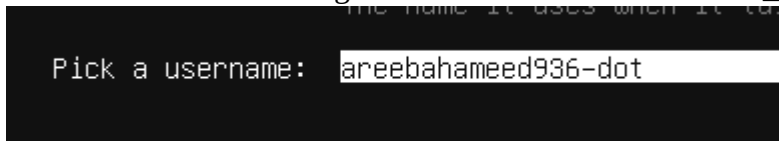
2. Enter ubuntu in the Your server's name field.
- Save a screenshot showing the server name as ubuntu_server_name.png.



Your servers name: ubuntu

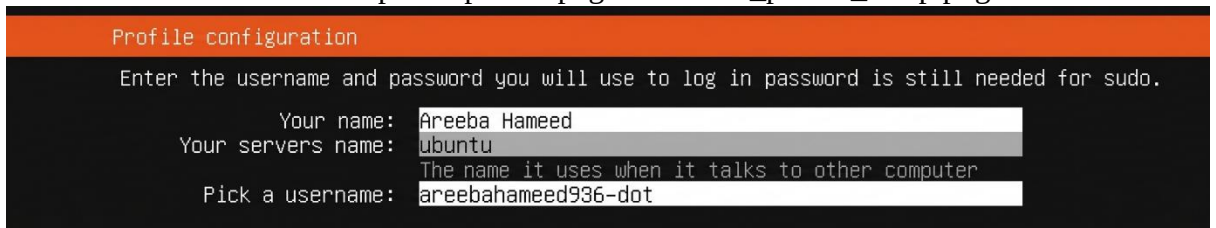
The name it uses when it talks to other computer

3. Enter your exact GitHub username in the Pick a username field.
- Save a screenshot showing the Ubuntu username as ubuntu_username.png.



Pick a username: areebahameed936-dot

4. Choose and confirm a secure password.
- Do not expose your password in any screenshot.
5. Review all profile fields before continuing.
- Save a screenshot of the completed profile page as ubuntu_profile_setup.png.



Profile configuration

Enter the username and password you will use to log in password is still needed for sudo.

Your name: Areeba Hameed

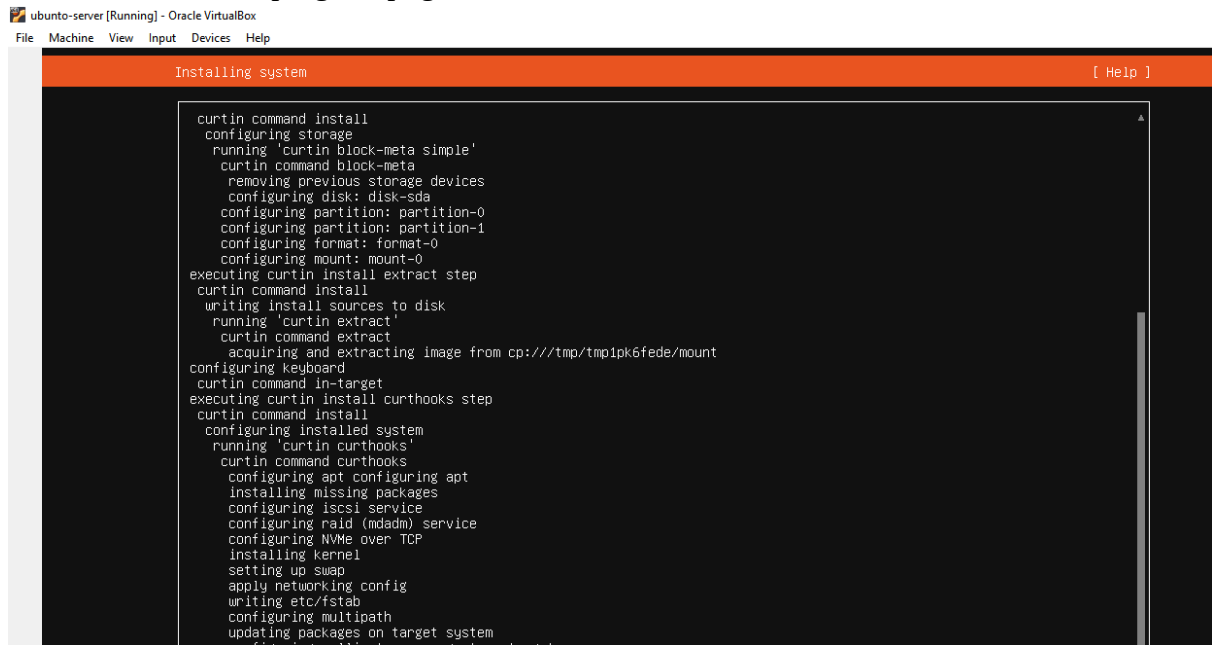
Your servers name: ubuntu

The name it uses when it talks to other computer

Pick a username: areebahameed936-dot

Task 14: Complete the Software Installation

1. Wait while Ubuntu Server installs the required software.
 - Save a screenshot showing the installation progress as `ubuntu_installation_progress.png`.



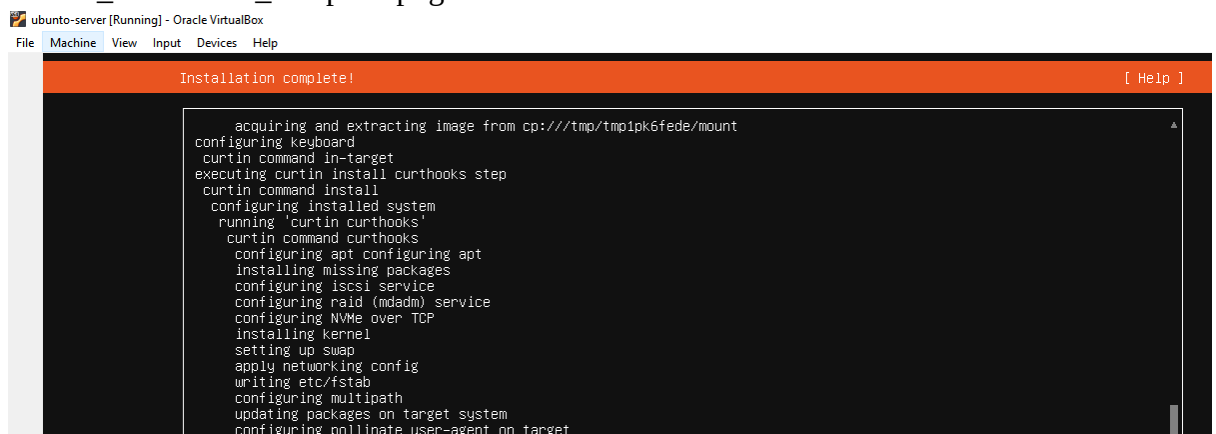
The screenshot shows a terminal window titled "Installing system" with a list of commands and their outputs. The commands are executed in a sequence, showing the progress of the installation. The terminal output includes:

```
curtin command install
configuring storage
running 'curtin block-meta simple'
curtin command block-meta
  removing previous storage devices
  configuring disk: disk-sda
  configuring partition: partition-0
  configuring partition: partition-1
  configuring format: format-0
  configuring mount: mount-0
executing curtin install extract step
curtin command install
  writing install sources to disk
  running 'curtin extract'
  curtin command extract
    acquiring and extracting image from cp:///tmp/tmp1pk6fede/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
    configuring apt configuring apt
    installing missing packages
    configuring iscsi service
    configuring raid (mdadm) service
    configuring NVMe over TCP
    installing kernel
    setting up swap
    apply networking config
    writing etc/fstab
    configuring multipath
    updating packages on target system
    configuring pollinate user-agent on target
```

2. Do not power off the virtual machine during the installation.
3. Wait until the installer reports that the installation is complete.

Task 15: Reboot and Verify the Ubuntu Identity

1. When the installation completes, select Reboot Now.
 - Save a screenshot of the installation-complete screen as `ubuntu_installation_complete.png`.



The screenshot shows a terminal window titled "Installation complete!" with a list of commands and their outputs. The commands are executed in a sequence, showing the progress of the installation. The terminal output includes:

```
acquiring and extracting image from cp:///tmp/tmp1pk6fede/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
    configuring apt configuring apt
    installing missing packages
    configuring iscsi service
    configuring raid (mdadm) service
    configuring NVMe over TCP
    installing kernel
    setting up swap
    apply networking config
    writing etc/fstab
    configuring multipath
    updating packages on target system
    configuring pollinate user-agent on target
```

2. Disconnect the Ubuntu ISO if VMware asks you to remove the installation media.
3. Allow the installed Ubuntu Server to start.
 - Save a screenshot showing the Ubuntu login screen as `ubuntu_login_screen.png`.

```
[ 33.576931] cloud-init[894]: | . =+.. 0|
[ 33.577688] cloud-init[894]: | S .0=0.. |
[ 33.584475] cloud-init[894]: | . 0.... |
[ 33.593416] cloud-init[894]: | . .+ .0|
[ 33.597285] cloud-init[894]: | . =.*|
[ 33.600438] cloud-init[894]: | 0=*.|
[ 33.607485] cloud-init[894]: +----- [SHA256] -----+

ubuntu login:
```

4. Sign in using the Ubuntu username and password configured in Task 13.
 - Save a screenshot showing the successful terminal login as `ubuntu_terminal_login.png`.

```
ubuntu login: areebahameed936-dot
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Oct 6 06:53:10 PM UTC 2026

System load:          0.08
Usage of /:            22.4% of 24.44GB
Memory usage:         14%
Swap usage:           0%
Processes:            97
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe63:b3a4

Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

areebahameed936-dot@ubuntu:~$
```

5. Verify that the terminal prompt follows this format:
`<github-username>@ubuntu:~$`
 - Save a clear screenshot showing the complete prompt as `ubuntu_identity_verified.png`.

```
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

areebahameed936-dot@ubuntu:~$
```

Task 16: Find the Ubuntu Server IP Address

1. Run the following command inside Ubuntu Server:
`ip addr`

- Save a screenshot showing the command and its output as `ubuntu_ip_addr_command.png`.

```
areebahameed936-dot@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:63:b3:a4 brd ff:ff:ff:ff:ff:ff
    altname enx08002763b3a4
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 85208sec preferred_lft 85208sec
    inet6 fd17:625c:f037:2:a00:27ff:fe63:b3a4/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86207sec preferred_lft 14207sec
    inet6 fe80::a00:27ff:fe63:b3a4/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
areebahameed936-dot@ubuntu:~$
```

2. Find the IPv4 address shown after `inet` for the active network interface.
- Do not use 127.0.0.1; it is the loopback address.
- Save a screenshot clearly showing the correct IPv4 address as `ubuntu_ip_address.png`.

```
altname enx08002763b3a4
inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope glob
    valid_lft 85208sec preferred_lft 85208sec
inet6 fd17:625c:f037:2:a00:27ff:fe63:b3a4/64 scope glo
    valid_lft 86207sec preferred_lft 14207sec
inet6 fe80::a00:27ff:fe63:b3a4/64 scope link proto ken
    valid_lft forever preferred_lft forever
areebahameed936-dot@ubuntu:~$
```

3. Keep your `<github-username>@ubuntu` prompt visible in the screenshot.

Task 17: Connect to Ubuntu Server from Windows

1. Keep the Ubuntu Server virtual machine powered on.
2. Open Command Prompt or PowerShell on Windows.
3. Run the following SSH command, replacing the placeholders with your Ubuntu username and server IP address:

```
ssh <github-username>@<ubuntu-server-ip>
```

- Save a screenshot showing the SSH command as `windows_ssh_command.png`.

```
Command Prompt - ssh areebahameed936-dot@127.0.0.1 -p 2222
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh areebahameed936-dot@127.0.0.1 -p 2222
The authenticity of host '[127.0.0.1]:2222 ([127.0.0.1]:2222)' can't be established.
ED25519 key fingerprint is SHA256:X4w0VA8tzDXLRSdcp0w/65oxkM376V0KNJQFcj519A.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

4. Type yes if you are asked to accept the server fingerprint.
- Save a screenshot showing the accepted fingerprint as `windows_ssh_fingerprint.png`.

```
areebahameed936-dot@ubuntu: ~
ED25519 key fingerprint is SHA256:X4w0VA8tzDXLRSdcp0w/65oxkM376V0KNJQFcXj519A.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[127.0.0.1]:2222' (ED25519) to the list of known hosts.
areebahameed936-dot@127.0.0.1's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Oct  6 07:12:14 PM UTC 2026
```

5. Enter your Ubuntu password and complete the remote login.
 - The password will not appear while you type; this is normal.
 - Do not expose your password in a screenshot.
 - Save a screenshot showing the successful SSH session as windows_ssh_login.png.

```
Warning: Permanently added '[127.0.0.1]:2222' (ED25519) to the list of
areebahameed936-dot@127.0.0.1's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Oct  6 07:12:14 PM UTC 2026

System load:          0.0
Usage of /:            22.5% of 24.44GB
Memory usage:         15%
Swap usage:           0%
Processes:            104
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe63:b3a4

Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

6. Verify that the remote prompt displays <github-username>@ubuntu.
 - Save a clear screenshot showing the remote prompt as windows_ssh_identity.png.

```
21 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

areebahameed936-dot@ubuntu:~$
```

If SSH is unavailable, install and start OpenSSH Server inside Ubuntu:

```
sudo apt update
```

```
sudo apt install -y openssh-server
```

```
sudo systemctl enable --now ssh
```